c) Reaction with thionyl chloride (Sulphonyl chloride)

Example

$$\underset{\text{Ethanol}}{CH_3CH_2OH} + SOCl_2 \xrightarrow{\text{Pyridine}} \underset{\text{Chloroethane}}{CH_3CH_2Cl} + SO_2\uparrow + HCl\uparrow$$

This reaction is known as **Darzen's halogenation**

2) From alkenes

Alkenes react with halogen acids (HCl, HBr, HI) to give haloalkane. The mode of addition follows Markovnikov's rule.

3) From alkanes

Alkanes react with halogens (Cl_2 or Br_2) in the presence of ultra violet light to form haloalkane. This reaction is a free radical substitution reaction and gives a mixture of mono, di or poly substituted haloalkane.

Example

Chlorination of methane gives different products which have differences in the boiling points. Hence, these can be separated by fractional distillation.

$$\underset{\text{Methane}}{CH_4} \xrightarrow[-\ HCl]{Cl_2\ /\ light} \underset{\text{Chloromethane}}{CH_3Cl} \xrightarrow[-\ HCl]{Cl_2} \underset{\text{Dichloromethane}}{CH_2Cl_2} \xrightarrow[-\ HCl]{Cl_2} \underset{\text{Trichloro methane}}{CHCl_3} \xrightarrow[-\ HCl]{Cl_2} \underset{\text{Tetrachloro methane}}{CCl_4}$$

4) Halogen exchange reactions

a) Finkelstein reaction

Chloro or bromoalkane on heating with a concentrated solution of sodium iodide in dry acetone gives iodo alkanes. This reaction is called Finkelstein reaction, (S_N2 reaction).

$$\underset{\text{Bromoethane}}{CH_3CH_2Br} + NaI \xrightarrow[\Delta]{\text{Acetone}} \underset{\text{Iodoethane}}{CH_3CH_2I} + NaBr$$

b) Swarts reaction

Chloro or bromo alkanes on heating with metallic fluorides like AgF, SbF_3 or Hg_2F_2 gives fluoro alkanes. This reactions is called Swarts reaction.

Example

$$\underset{\text{Bromo ethane}}{CH_3CH_2Br} + AgF \xrightarrow{\Delta} \underset{\text{Fluoro ethane}}{CH_3CH_2F} + AgBr$$

5) From silver salts of fatty acids (Hunsdiccker reaction)

Silver salts of fatty acids when refluxed with bromine in CCl_4 gives bromo alkane

$$\underset{\text{Silver propionate}}{CH_3CH_2COOAg} + Br_2 \xrightarrow[\text{reflux}]{CCl_4} \underset{\text{Bromo ethane}}{CH_3CH_2Br} + CO_2 + AgBr$$

Unit 14 - jagan.indd 231 22-01-2021 13:04:57